Masoud Hemmatpour

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Nationality: Iranian

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Education

- **Simula Lab** — Oslo, Norway
  Postdoctoral Fellow — *Sep. 2021 – Current*
  - **Project**
    - In-network computing in HPC systems
- **Cisco systems** — Paris, France
  Postdoctoral Fellow — *Jan. 2019 – Jan. 2020*
  - **Project**
    - Control and management plane in cloud-native environment
- **Politecnico di Torino** — Torino, Italy
  PhD in computer engineering — *Nov. 2015 – Sept. 2019*
  - **Thesis**
    - RDMA-enabled in-memory key-value store over InfiniBand protocol
- **Politecnico di Torino** — Torino, Italy
  MSc in communications and computer networks engineering — *Oct. 2012 – Oct. 2015*
  - **Thesis**
    - Read-copy update synchronization technique improvement by hardware message passing
      Research project at École Politechnique Fédérale de Lausanne (EPFL)
- **Azad university of najaf abad** — Najaf abad, Iran
  Bachelor of Science, Computer Software Engineering — *2005 – 2008*
  - **Thesis**
    - Implementing bulk SMS (Short Message Service) sender with AT commands in Delphi programming
- **Bu-Ali Sina University** — Hamedan, Iran
  Association Degree, Computer Software — *2002 – 2004*
  - **Thesis**
    - Implementing answering machine with TAPI , Delphi programming

Selected publications

- Communicating Efficiently on Cluster-Based Remote Direct Memory Access (RDMA) over InfiniBand Protocol
  Journal of Applied Sciences, 2018
- Analyzing In-Memory NoSQL Landscape
  IEEE Transactions on Knowledge and Data Engineering, 2020
- Kanzi: A distributed In-memory Key-Value store
  Middleware, Trento, Italy, 2016
- A kinematic-based indoor fall surveillance
  IEEE Access, under review
- DIIG : A Distributed Industrial IoT Gateway
  COMPSAC, Torino, Italy, 2017
- Polynomial Classification model for real-time fall prediction system
  COMPSAC, Torino, Italy, 2017

- A baseline walking dataset exploiting accelerometer and gyroscope for fall prediction and prevention systems
  BodyNets, Torino, Italy, 2016
- Nonlinear Predictive Threshold Model for Real-Time Abnormal Gait Detection
  Journal of Healthcare Engineering, 2018
- Internet of Things for fall prediction and prevention
  Journal of Computational Methods in Sciences and Engineering, 2018
- Urban Dust Monitoring From Ground Level to Last Floor
  ICMU, Toyama, Japan, 2017
- Analysis and Optimization of Synchronization Algorithms for Multicore Architectures
  RENS workshop Tallinn, Estonia, 2016

Work experience

- Network management (Linux Foundation Collaborative Project in Cisco Systems)
  - Designing and implementing a Yet Another Next Generation (YANG) model to enable VPP to communicate with remote controllers such as OpenDaylight (ODL) and Network Services Orchestrator (NSO) in order to receive telemetry and push configuration.
  - Enabling VPP to communicate with a network routing software suite such as FRRouting (FRR).
- Remote Direct Memory Access (RDMA) (Research project Politecnico di Torino & T.J. Watson IBM research center)
  - Investigating performance challenges of RDMA operations in InfiniBand based clusters.
  - Designing and implementing an RDMA enabled in-memory key-value store.
- Health care system (OPportunities for active and healthy LONgevity (OPLON) project)
  - Designing and implementing a holistic approach to detect an abnormal gait in order to avoid an unintentional fall and in case of a fall detection reduces injuries and notifies care givers.
- Fault injection (Research Project Politecnico di Torino & University of Montpellier)
  - Designing and implementing a framework to automatically inject faults into a real-time operating system (i.e., FreeRTOS) for STM32 Discovery boards.
- Linux kernel scheduler optimization (Research Project at Politecnico di Torino)
  - Introducing new parameter (i.e., the number of data cache misses) at the given time slice to the Completely Fair Scheduler (CFS). In this way, the scheduler can penalizes processes which waste CPU cycles to handle cache miss instead of the main task.
- Synchronization algorithm on multi core architecture (Research Project at Politecnico di Torino)
  - Implementing a framework to evaluate different synchronization methods including non-blocking algorithms.
- Urban air quality (Cyclair project Torino Living Lab)
  - Investigating the relevance of height in dust monitoring systems based on Libelium Waspmote connected to a dust sensor OPC-N2.
- System-on-a-chip (Research project Politecnico di Torino & École polytechnique fédérale de Lausanne (EPFL))
  - Exploiting hardware message passing feature in system-on-a-chip CPU (i.e., TILE-Gx36) in order to enhance the performnce of the non-blocking synchronization algorithm.
- Co-founder Fanavaran Pasargad J, Linux Professional Institute (LPI) partner

  Linux Trainer
  - Isfahan university of technology
  - Isfahan university
  - Isfahan regional electric company
  - Many other companies and universities.

- 3D Simulation
  - Designing and implementing a flight simulator through an OpenGL based game engine (Glscene).

Visiting programs

École Polytechnique Fédérale de Lausanne (EPFL)	*Jun. 2015*
Purdue University	*Feb. 2016*
University of California Davis (UC DAVIS)	*Feb. 2017*

Awards

Quality Award Politecnico di Torino 2017

Quality Award Politecnico di Torino 2018

Best Dissertation Award Politecnico di Torino 2019

Interests

High Performance Computing (HPC): Exploiting modern hardware such as System-on-chip, Remote Direct memory Access (RDMA), 3D XPoint memory, and non-volatile memory express (NVMe) to enhance the performance of existing data analysis.

Operating system: designing novel synchronization mechanism, Inter Process Communication (IPC), and real-time scheduler.

Performance evaluation: Performance evaluation of parallel and distributed systems.

Big data and artificial intelligence: Analyzing Big data to apply supervised and unsupervised learning.

Biomedical and health science: biomedical computing, assistive robotics, neuroscience.

Language skills

Farsi: Farsi is my mother tongue.

English: IELTS international certification (overall 6.0).

Italian: Intermediate (level B2).

References

Maurizio Rebaudengo

Full professor at the department of computer and control engineering Politecnico di Torino, Italy
Web site: http://www.dauin.polito.it/personale/scheda/(nominativo)/maurizio.rebaudengo
Email:maurizio.rebaudengo@polito.it

Mohammad Sadoghi

Assistant professor at the department of computer science University of California Davis, USA
Web site: https://faculty.engineering.ucdavis.edu/sadoghi/
Email: mo.sadoghi@expolab.org

Matteo Sonza Reorda

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